

Manufacturing Functional Muscle Tissues to Test Mobility-Restoring Therapies

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Abstract—Neuromuscular diseases such as amyotrophic lateral sclerosis, sarcopenia, and muscular dystrophies affect millions of people worldwide and severely reduce mobility, independence, and quality of life. In many cases, existing therapies fail to restore force generation, creating a major need for scalable experimental platforms compatible with high-throughput screening to identify force-restoring treatments. The work presented here advances STAMP (Simple Templating of Actuators via Micro Topographical Patterning), a hydrogel-based skeletal muscle patterning platform that enables aligned, contractile two dimensional muscle tissues. Building on prior work, several additions were developed to move the platform toward high-throughput functional screening. Cross-shaped force markers were incorporated to provide persistent visual reference features for downstream force analysis. The platform was adapted to a 96-well plate format to support higher throughput drug screening, and an alignment grid was designed to improve assembly speed, consistency, and reproducibility. In parallel, we developed a STAMP compatible with an electrical stimulation device allowing for exercise-like muscle stimulation at the 6-well plate scale. Ultimately, we successfully demonstrated improved marker visibility, successful large-scale hydrogel patterning, and reduced groove angle standard deviation with the alignment grid, suggesting more accurate and reproducible tissue patterning and downstream force measurements. These findings establish a foundation for large-scale, simple production of aligned and contractile muscle tissues, paving the way for future studies of mobility-restoring drugs, including the initiation of therapeutic screening for Duchenne muscular dystrophy (DMD) at the Massachusetts General Hospital (MGH).

Index Terms—skeletal muscle engineering, neuromuscular disease, drug screening, hydrogel patterning, microtopography, high-throughput screening, electrical stimulation

I. INTRODUCTION

Neuromuscular diseases, including amyotrophic lateral sclerosis, muscular dystrophies, sarcopenia, and rare genetic muscle disorders, affect approximately 15 million people worldwide and can progressively impair walking, standing, swallowing, and breathing [1]. Because these disorders ultimately diminish skeletal muscle force generation, there remains a strong need for experimental apparatuses that can measure functional improvement outside the living body in a non-invasive manner through *in vitro* models. Existing drugs frequently fail to restore contractile performance, creating substantial demand for platforms that can identify mobility-restoring therapies with improved predictive value.

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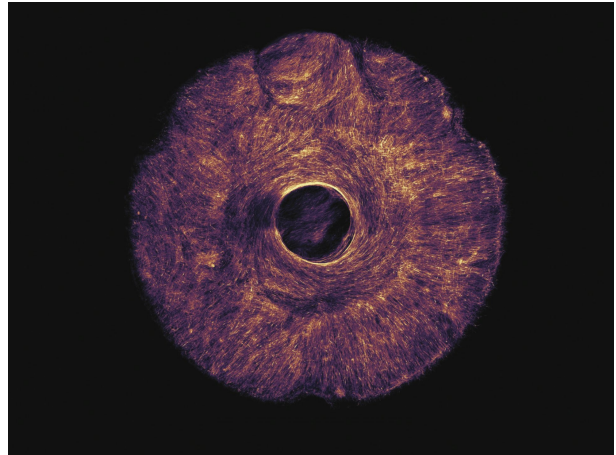


Fig. 1. Example of STAMPed muscle tissue with microtopography inspired from iris architecture [2].

Engineered skeletal muscle has emerged as an important *in vitro* model for disease modeling, therapeutic screening, and biohybrid robotics. Traditional three dimensional muscle tissues formed between compliant posts or cantilevers have proven useful because they generate measurable forces and reproduce native muscle fiber alignment. However, these systems often require complex fabrication, are relatively low throughput, and are less compatible with high-content imaging. By contrast, two dimensional skeletal muscle cultures are more scalable and imaging friendly, but they typically suffer from inconsistent pattern alignment, disorganized contractions, and delamination from rigid substrates.

The Raman Lab addressed these limitations through Simple Templating of Actuators via Micro Topographical Patterning (STAMP), a one step method that patterns microscale grooves into compliant hydrogels using two-photon lithography printed STAMPs [2]. This approach combines advantages of both two dimensional and three dimensional muscle systems, namely streamlined imaging, muscle alignment, and anisotropic muscular contraction. The original platform demonstrated that aligned mouse and human skeletal muscle tissues could be fabricated reproducibly across multiple groove geometries without negatively affecting maturation or twitch magnitude [2]. More complex architectures, including multi oriented iris inspired actuators (Fig. 1), were also demonstrated, highlighting the versatility of the STAMP method for guiding tissue level motion [2].

However, this early work was limited to low throughput

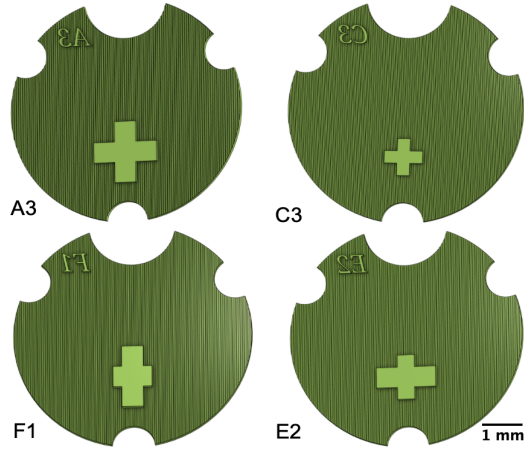


Fig. 2. Selected CAD renderings of STAMPs with different cross geometries used to pinpoint the optimal dimensions [3].

on the 24-well plate with only 1 to 2 wells patterned at once and manual assembly of the 3D-printed parts, which resulted in well-to-well groove direction variability. As such, the present work extends this platform toward a screening-oriented system designed to identify therapies that restore muscle function. Several engineering additions were developed to meet this goal: cross-shaped reference markers, a 96-well plate architecture for four times higher throughput production, an alignment grid to optimize the assembly process and reduce groove direction variability, and a stamp compatible with a 6-well plate electrical stimulation apparatus to study exercise-like activation as a complementary therapeutic modality. In addition, preliminary studies have been run with DMD cells, showing STAMP's potential for generating high-throughput disease models.

II. METHODS

A. Design of Cross Marked STAMP Geometries

To facilitate downstream force analysis, we designed STAMPs with cross-shaped reference markers with varying dimensions and explored whether recessed or protruding cross geometries produced higher patterning fidelity. The grooves were designed at $25\ \mu\text{m}$ depth and width, consistent with the size scale previously shown to support muscle alignment on STAMPed fibrin [2]. The cross markers were introduced as visual reference features intended to remain identifiable before and after cell seeding, thereby allowing tissue displacement to be mapped across imaging sessions.

Six STAMP variants, denoted A through F, were fabricated with differing cross geometries and three height conditions of 60, 90, and $120\ \mu\text{m}$ as shown in Fig. 2. Cross features ranged from 30 to $120\ \mu\text{m}$ in width and 900 to $1500\ \mu\text{m}$ in length. These features were designed to create localized topographical signatures within the hydrogel that would remain visible following muscle growth. Fibrin substrates were used, as prior studies have shown they support greater maximal contraction than alternative hydrogels such as collagen [4].

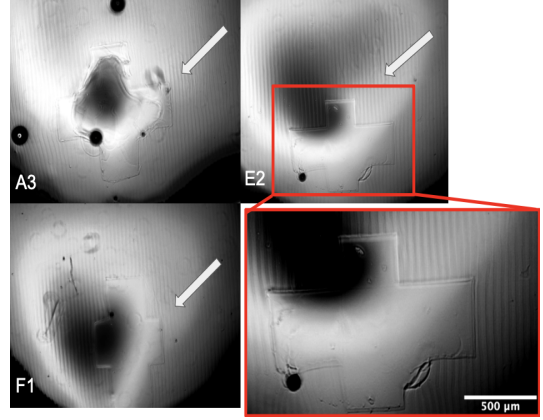


Fig. 3. Corresponding hydrogel microscopy imaging for STAMPs in Fig. 2, showcasing clear cross features embedded into the hydrogel.

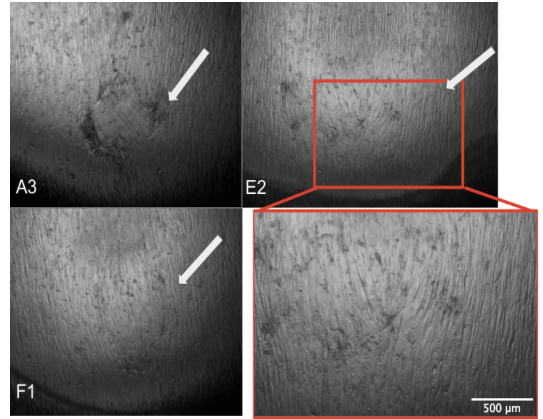


Fig. 4. Corresponding hydrogel microscopy imaging with cells for STAMPs in Fig. 2, illustrating smoothed out cross features that lose shape fidelity.

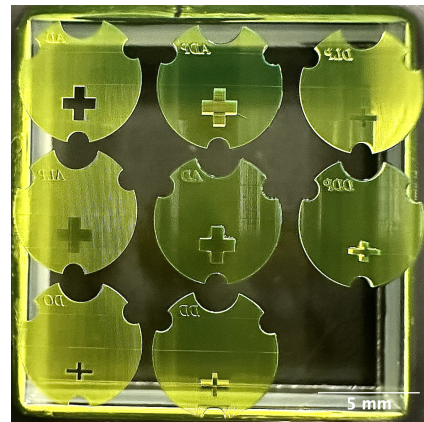


Fig. 5. Revised STAMP designs with varying cross geometries, extrusions and cavities.

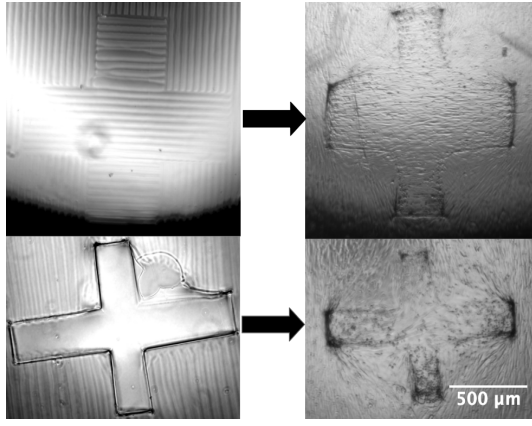


Fig. 6. Hydrogel patterning from STAMPs in Fig. 5 pre and post cell seeding, showing maintained fidelity in the cross feature.

B. Development and Fabrication of the Stamp and Holder compatible with 96-well plates

STAMP designs were fabricated using high resolution two-photon lithography, enabling rapid iteration across multiple geometries. To support parallel patterning at a larger scale, the individual STAMPs were mounted onto custom holders. The holder frame was adapted toward a 12×8 arrangement corresponding to a 96-well plate format. In conjunction with multichannel pipetting, this setup enabled mass hydrogel patterning across the plate within minutes.

C. Hydrogel Optimization

Because hydrogel fidelity is critical for preserving microtopographical features, the composition of the fibrin substrate was optimized to balance compliance with pattern retention. Hydrogels were formed using fibrinogen and thrombin, following the general strategy used in related Raman Lab work on contractile skeletal muscle culture [2], [3]. Fibrinogen concentration was tuned to control gel stiffness, while thrombin concentration governed crosslinking kinetics. An optimized fibrinogen concentration of 12 mg/mL and an approximately 30:1 fibrinogen to thrombin ratio produced a compliant hydrogel layer that retained both small alignment grooves and larger marker indentations as seen in Fig. 3.

D. Cell Seeding and Early Tissue Culture

After hydrogel patterning, C2C12 myoblasts were seeded onto the patterned hydrogel substrates and expanded in growth medium until confluence was reached. Growth medium was then replaced with differentiation medium, promoting the fusion of myoblasts into differentiated muscle fibers. The differentiated cells were then assessed after cell seeding and during subsequent culture to determine whether cross-shaped reference features remained distinguishable as the tissue matured. Given that cross fidelity diminished as shown in Fig. 4, these observations were used to guide iterative redesign of the STAMP geometry (Fig. 5).

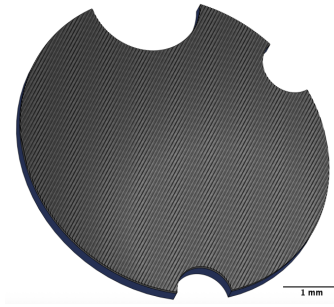


Fig. 7. STAMP CAD model displaying asymmetrical alignment notches for ease of identifying STAMP orientation.

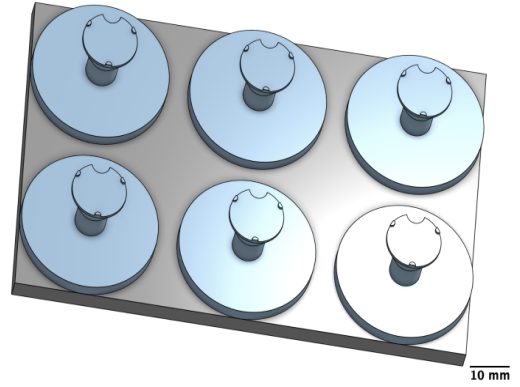


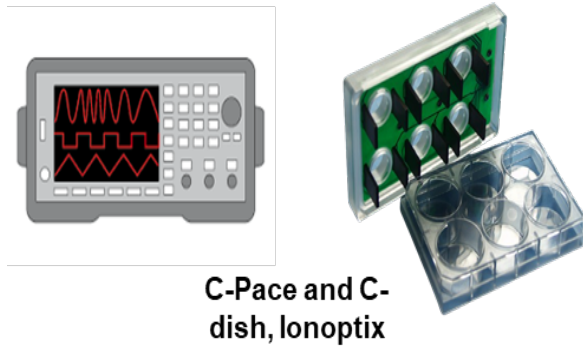
Fig. 8. Custom 6-well plate STAMP holder compatible with the electrical stimulation platform in Fig. 9

E. Alignment Grid Development

In addition to STAMP holders, an alignment grid was made to ensure highly reproducible groove alignment. Prior to this development, assembly relied on tape and tweezers to mount individual STAMPs, introducing inconsistency in stamp alignment and reducing reproducibility across wells. The alignment grid was designed to allow STAMPs to be placed into a defined array and then attached simultaneously by pressing the tape-coated holder onto the grid. This approach was intended to increase throughput while also preserving alignment consistency across the full plate. Furthermore, asymmetry was incorporated into the STAMP to allow for easy identification of the side with microtopological patterning (Fig. 7).

F. Six-Well Electrical Stimulation Platform

A complementary 6-well STAMP platform (Fig. 8), compatible with a commercially-available electrical stimulation platform was also developed to investigate repeated stimulation of engineered muscle tissues, mimicking the effects of regular exercise. This array consisted of STAMPs with a diameter of 6.75 mm, one millimeter smaller than the original 35 mm diameter of the 6-well plate, which were designed to fit into custom laser-cut tiles (Fig. 10), confining the cells to a defined growth area while maintaining space for the electrodes of the electrical stimulation platform (Fig. 9).



C-Pace and C-dish, lonoptix

Fig. 9. Muscle exercising electrical stimulation device compatible with the 6-well plate. Courtesy of Brandon Rios, Raman Lab.

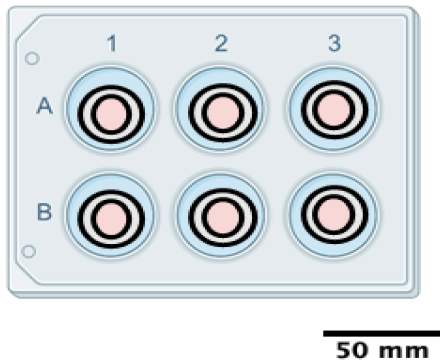


Fig. 10. Top down view of the 6-well plate with custom acrylic tiles in the wells serving as boundaries for the muscle tissue growth area while accommodating the electrodes shown in Fig. 8. Courtesy of Brandon Rios, Raman Lab.

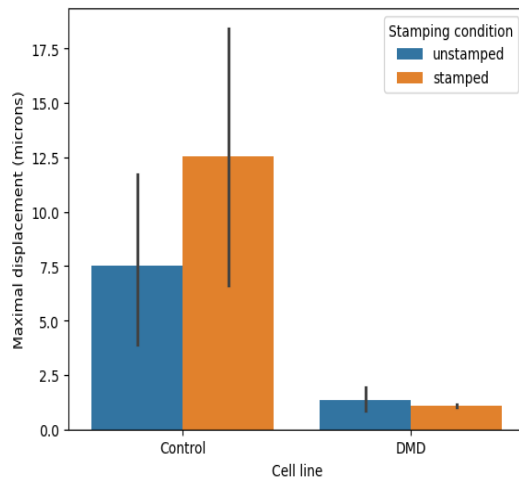


Fig. 11. Graph showing maximal muscle displacement from electrical stimulation, comparing STAMPed and unSTAMPed cells.

G. DMD Cell Integration and Stimulation Assays

Human immortalized myoblasts, including both healthy controls and DMD lines, were obtained through collaborators at MGH and cultured using protocols similar to C2C12 cells. To assess functional activation, engineered muscle tissues were subjected to electrical stimulation using titanium wires as electrodes, connected to a function generator sending square pulses at 1 Hz frequency, with 20% duty cycle and 10 V amplitude (Fig. 11). As a higher-throughput alternative, we also tested chemical stimulation with 10 μ M acetylcholine, the neurotransmitter naturally secreted by motor neurons to trigger muscle contraction. Muscle contraction was recorded at 10X magnification and 30 frames per second, using a camera mounted on a brightfield microscope. A custom tracking algorithm was used to quantify the magnitude and frequency of muscle displacement [4].

III. RESULTS

A. Hydrogel Patterning Preserved Both Grooves and Marker Features

The optimized fibrin hydrogel retained both the small alignment grooves and the larger cross-shaped marker indentations after patterning as depicted in Fig. 6. Early microscopy of the patterned gels confirmed that the STAMP features transferred successfully into the hydrogel surface, indicating that the selected gel composition was sufficient to preserve microscale and mesoscale topography simultaneously.

B. Marker Visibility Depended More Strongly on Depth Than Lateral Size

After C2C12 seeding and several days of early maturation, the visibility of the cross-shaped markers varied substantially across designs. Contrary to the initial expectation that larger crosses would produce clearer reference features, marker clarity was found to depend more strongly on feature depth than on lateral width. Several smaller cross designs generated more distinct outlines than larger ones, suggesting that hydrogel deformation and edge collapse limited the usefulness of simply enlarging the cross footprint.

The underlying reason appeared to be the elasticity of the fibrin hydrogel. Rather than preserving sharply defined ridges, the gel formed more gradual, sloped contours around the patterned features. As a result, shallow protrusions became less distinguishable after cell growth, whereas deeper features retained greater visual contrast. These observations motivated a redesign of the STAMP markers toward increased depth and stronger local topographical contrast.

C. Redesigned STAMPs Improved Post Seeding Marker Contrast

Based on the initial results, the STAMP geometries were redesigned to include deeper cross markers, perpendicular alignment grooves at the surface of the crosses, and recessed cavities in place of protruding features (Fig. 6). These modifications produced visibly improved hydrogel patterns both

before and after cell seeding. Marker outlines remained apparent after early cell maturation, suggesting that the redesigned topography is better suited for downstream force analysis than the initial generation of cross markers.

D. Successful 96-Well Plate Scale Up

The STAMP holder design enabled simultaneous patterning across a 96-well plate layout (Fig. 12), demonstrating that the platform can be moved from proof of concept fabrication toward higher throughput manufacturing. Instead of patterning individual substrates sequentially, the 12×8 holder architecture allowed an entire 96-well plate, commonly used in the pharmaceutical industry, to be patterned in a single step. This scale up is a key requirement for future drug screenings, where many compounds, doses, and biological replicates must be tested side by side.

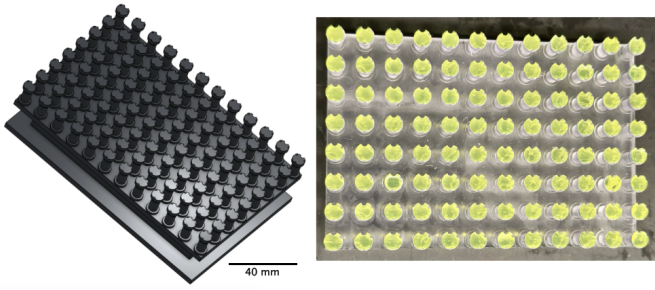


Fig. 12. 12×8 STAMP holder for simultaneous patterning in a 96-well plate format.

E. The Alignment Grid Reduced Variability in Pattern Orientation

The use of the alignment grid (Fig. 13) reduced variability in groove angle across patterned wells compared to manual assembly. This lower standard deviation in groove angle indicates that the grid improves pattern consistency and preserves alignment across the array, allowing for more accurate force analysis. Fig. 14 compares the mean groove angle with standard deviation for manual and high-throughput assembly. The manually assembled samples feature a noticeably wider spread in angle measurements (approximately 4 times larger standard deviation compared to high-throughput assembly), whereas the grid-aligned samples show a tighter clustering, indicating improved consistency.

IV. DISCUSSION

The results presented here extend STAMP from a proof-of-concept patterning method into a more practically scalable platform for functional neuromuscular research. Prior work established that STAMP can reproducibly align mouse and human skeletal muscle on compliant fibrin hydrogels while maintaining contractility and compatibility with imaging [2]. The current work builds on that foundation by addressing several bottlenecks that become critical when the goal shifts

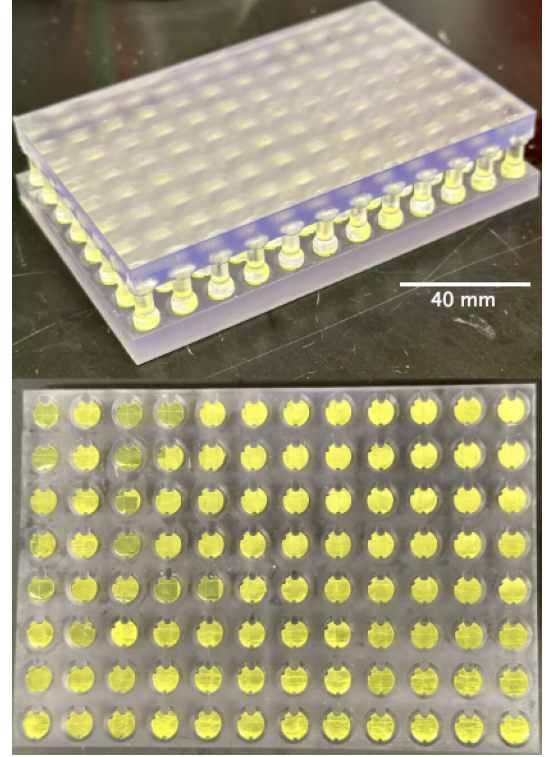


Fig. 13. Custom 96-well STAMP alignment grid enabling simultaneous high-throughput hydrogel patterning. (Top) Perspective view of the assembled device. (Bottom) Top-down view of the 12×8 grid.

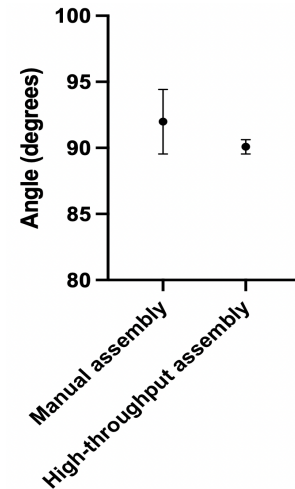


Fig. 14. Graph showing the mean groove angle with standard deviation for manual vs. high-throughput assembly. The standard deviation for manual assembly is approximately 5 times larger than that of the high-throughput assembly.

from demonstrating aligned muscle tissues to deploying them in a screening workflow.

One challenge in force based assays is the availability of visual landmarks that remain detectable across culture and imaging conditions over multiple days. The cross-shaped marker study showed that merely increasing the lateral size of a feature is insufficient when the underlying hydrogel is compliant and prone to spontaneous collapse or active re-modeling by growing cells. Instead, the vertical topographical contrast appears to be a more important design variable for cross marker visibility.

Another important advance is the move toward 96-well plate fabrication. A major limitation of many engineered muscle platforms is that they remain too labor intensive for routine compound screening. The ability to pattern many wells in parallel is therefore essential if functional readouts are to be used at scale. However, throughput alone is not sufficient; reproducibility must also be preserved. The alignment grid addresses this second requirement by replacing error-prone manual placement with a more controlled assembly strategy. The resulting reduction in groove angle standard deviation suggests that the new assembly process improves assay consistency at the topographical level, which should, in turn, improve consistency at the biological and functional levels.

The integration of a commercially available six-well electrical stimulation system expands the functionality of the platform. While the 96-well format enables high-throughput drug screening, many neuromuscular diseases are not purely pharmacological in nature, but are also strongly influenced by activation patterns and contraction dynamics. Importantly, the effects of exercise in these pathologies remain debated, as certain regimes can improve outcomes while others may exacerbate muscle damage. Given this, such a platform can be used to identify beneficial or detrimental stimulation regimes (e.g., frequency and intensity), evaluate drugs in the context of exercise-induced damage, and explore potential therapeutic benefits of targeted stimulation protocols. However, given that electrical stimulation is difficult to scale to high-throughput formats, we also explored chemical stimulation using acetylcholine. This approach enables rapid and parallel activation of tissues without external hardware, making it more compatible with 96-well plate screening. Comparing responses between control and DMD-derived tissues under these stimulation modalities provides a framework for identifying conditions that restore contractile performance.

V. CONCLUSION

A series of engineering advances were developed to move STAMP toward a high-throughput and more informative skeletal muscle assay platform. Cross-shaped reference markers were introduced to enable force tracking, and iterative redesign showed that deeper, higher contrast features improve post seeding visibility more effectively than simply increasing marker width. The platform was successfully adapted toward 96-well plate hydrogel patterning through custom holder designs, establishing the basis for larger scale manufacturing of

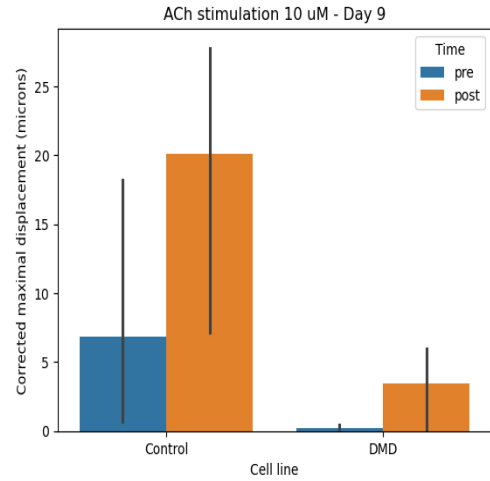


Fig. 15. Muscle maximal displacement with 10 μ M acetylcholine plotted against Control and DMD muscle experimental groups for both pre and post stimulation. This illustrates a nearly threefold increase in muscle contraction displacement between pre and post acetylcholine addition.

patterned muscle tissues. In addition, a modular alignment grid reduced groove angle variability across the array, improving reproducibility and strengthening the basis for accurate force analysis. Finally, a complementary stamp compatible with a commercially-available six-well electrical stimulation platform expanded the system beyond drug screening alone and enabled studying exercise-like interventions.

Together, these results position STAMP as more than a microtopographical patterning method. The platform now functions as an integrated experimental system for identifying therapies that restore muscle function, with applications in drug discovery, disease modeling, and engineered tissue actuation.

VI. FUTURE STUDIES

Several next steps follow directly from this work, spanning platform development, quantitative readouts, and biological application. First, further automation of STAMP assembly remains a key objective. While the alignment grid substantially improves reproducibility and throughput, full automation would minimize user dependence and enable robust industrial-scale screening.

Second, the introduction of cross-shaped reference markers opens up the possibility of correlating functional outputs with underlying molecular features at the single fiber resolution, including the expression of specific proteins (visualized by antibody staining) and RNA (assessed by spatial transcriptomics). For example, local variations in contractile strength could be directly linked to differences in myosin expression or RNA signatures within specific regions of a construct.

A major direction for future work is the application of this platform to disease modeling, particularly Duchenne Muscular Dystrophy (DMD). Our preliminary results demonstrate that healthy muscle tissues exhibit stronger contractile behavior than dystrophic counterparts, establishing a functional baseline

(Fig. 11). In this context, alternative stimulation modalities should be explored, since the current electrical stimulation setup involves manually connecting the leads to the function generator to trigger the signal, which is not feasible in a high-throughput context. In addition, commercial electrical stimulation plates are not readily available in a 96-well plate format. Triggering muscle contraction through chemical stimulation, owing to its compatibility with automated liquid handlers, is therefore more adapted to compound screening in a pharmaceutical context. We ran preliminary experiments in a few wells of a 24-well plate, using chemical stimulation with acetylcholine, an inherent compound that motor neurons secrete to activate muscles. Acetylcholine stimulation produced strong contractile responses in both healthy and DMD cells (Fig. 15). Future work will involve testing this compound in a 96-well plate format, in combination with high-content imaging setups.

Ultimately, our platform will be used to systematically evaluate emerging therapies by comparing treated and untreated DMD-derived tissues against healthy controls. This enables direct assessment of whether candidate therapies restore contractile function toward physiological levels. Ongoing collaborations, including work with the Rengarajan Lab at MGH, aim to leverage this system for high-throughput therapeutic screening across diverse conditions.

Finally, the 6-well plate platform enables the study of disease-relevant mechanical stress. Because repeated contraction can exacerbate muscle damage in DMD, the ability to apply controlled, repeated stimulation provides a means to model exercise-induced damage in vitro. Comparing stimulated and non-stimulated conditions may reveal differential sensitivity to mechanical load and provide a more physiologically relevant framework for evaluating therapeutic efficacy.

More broadly, while this work focuses on skeletal muscle, the same patterning, assembly, and analysis principles could be extended to other contractile tissues, including cardiac and smooth muscle, thereby expanding the range of applications for STAMP-based manufacturing and functional analysis.

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